SOFT SENSOR APPLICATION ON DEBUTANIZER COLUMN

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Outline

Introduction & Background

Methodology

Exploratory Data Analysis

Results & Evaluation

Conclusion

Debutanizer Column

- A debutanizer column is used to separate light hydrocarbons (C4) from heavier fractions.
- Separation occurs due to differences in boiling points of hydrocarbons.
- Feed typically comes from catalytic cracking or reforming units.
- Lighter hydrocarbons leave from the top of the column.
- Heavier hydrocarbons are collected from the bottom.
- A condenser at the top provides reflux to enhance separation.
- A reboiler at the bottom supplies heat for vaporization.

Debutanizer Column Schematic

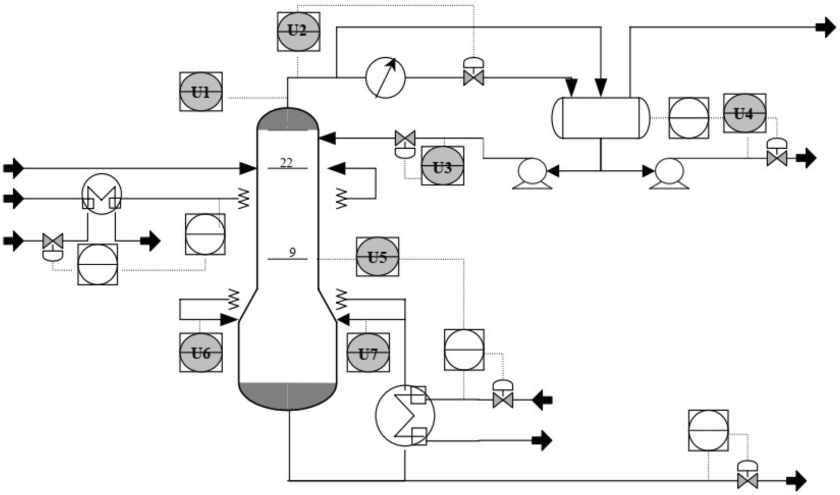


Figure: Schematic of the Debutanizer Column

Description of the input variables

Input Variables	Description
u1	Top temperature
u2	Top pressure
u3	Reflux flow
u4	Flow to next process
u5	Sixth tray temperature
u6	Bottom temperature
u7	Bottom temperature

Problem Statement

- Product composition is a key quality variable in distillation processes.
- In debutanizer columns, monitoring butane composition is essential.
- Direct composition measurement using analyzers is slow and costly.
- Gas chromatography introduces significant measurement delays.
- These delays reduce the ability to perform real-time monitoring.
- **Accurate and fast estimation of composition variables is therefore required.**
- **Soft sensors can estimate composition using easily measurable process variables.**

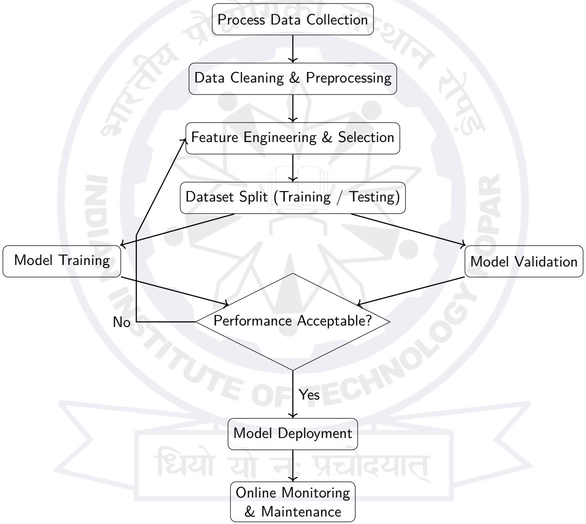
Classification of Soft Sensors

- First-principles (model-based) rely on fundamental process equations.
- Data-driven use historical plant data to build predictive models.
- Hybrid combines mechanistic models with machine learning.
- Static assume no time dependency; dynamic consider time-dependent behavior.
- Linear soft sensors: regression, PCA.
- Nonlinear soft sensors: advanced machine learning techniques.
- Choice depends on process complexity and data availability.

Disadvantages of Data-Driven Soft Sensors

- Require large amounts of high-quality historical data.
- Models may fail when operating conditions change significantly.
- Periodic retraining is required.
- Difficult to interpret compared to first-principles models.

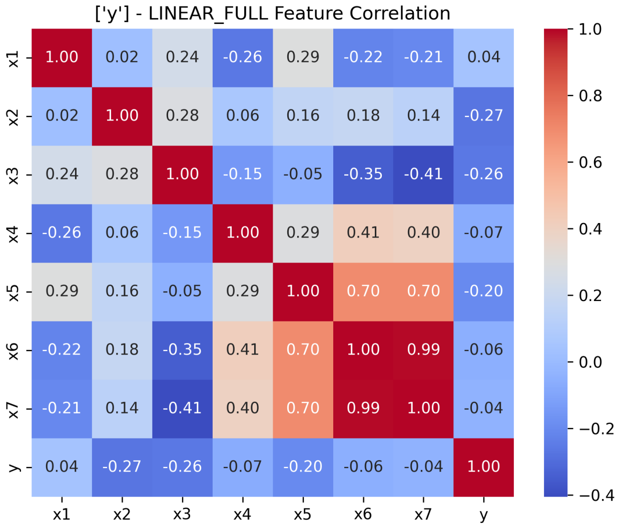
Soft Sensor Development Flowchart



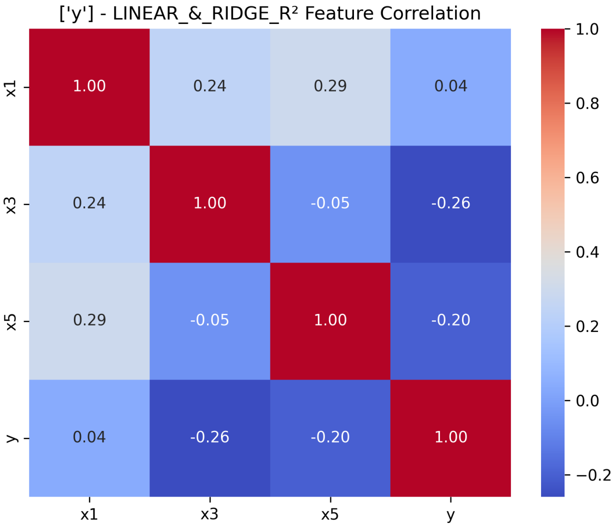
Soft Sensor Models Used

- Linear Regression – fits a line minimizing squared errors.
- Lasso Regression (L1) – performs feature selection automatically.
- Ridge Regression (L2) – stabilizes models with correlated features.
- Polynomial Regression (2nd order) – captures nonlinear relationships.
- Just-In-Time Learning (JITL) – builds a local model for each prediction.
- Lasso + Polynomial – combines feature selection and nonlinearity.
- Ridge + Polynomial – regularized nonlinear modeling for correlated features.

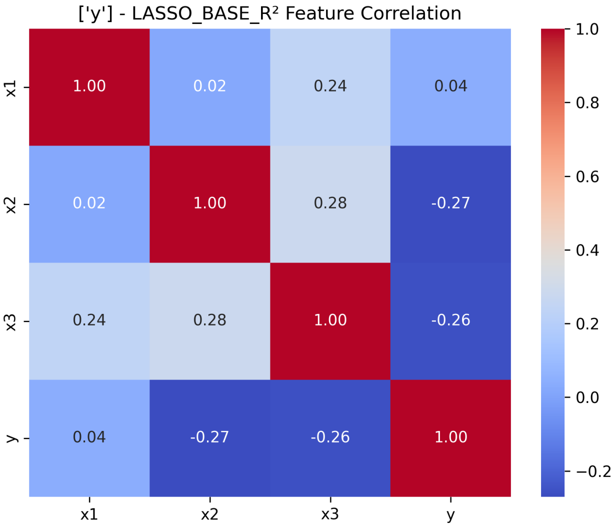
Correlation Analysis



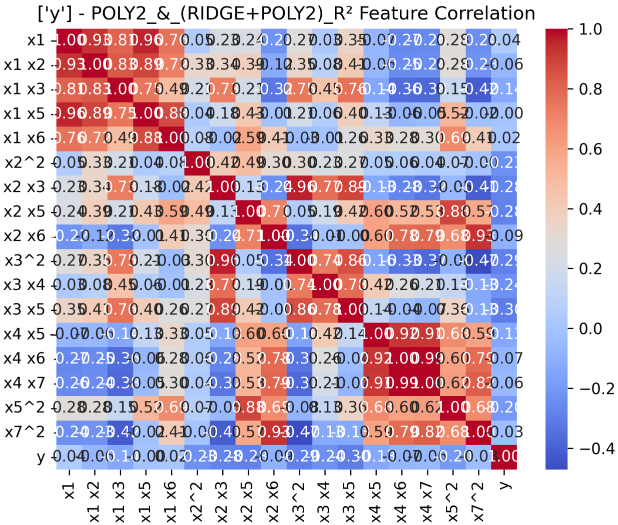
Selected Features Correlation (Linear & Ridge)



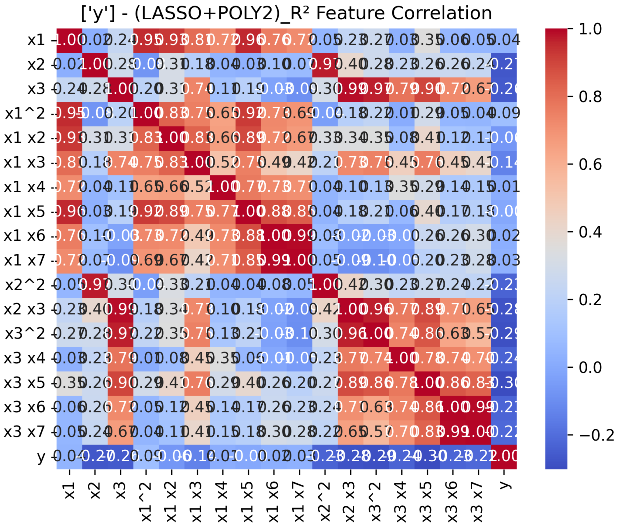
Selected Features Correlation (Lasso)



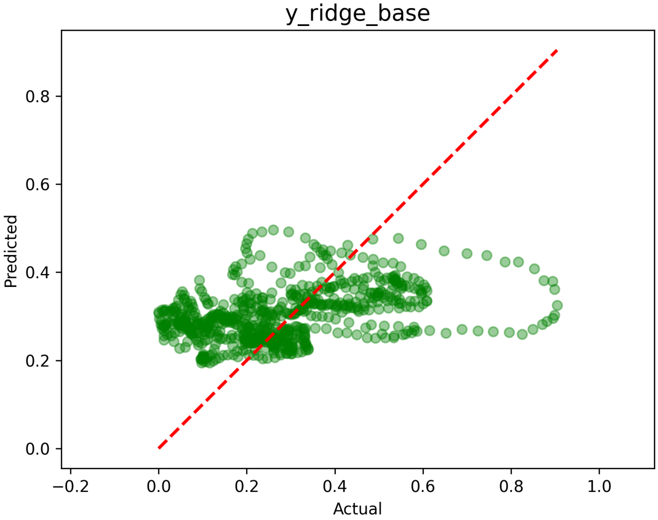
Selected Features Correlation (Polynomial(2) & (Ridge + Poly2))



Selected Features Correlation (Lasso + Poly2)



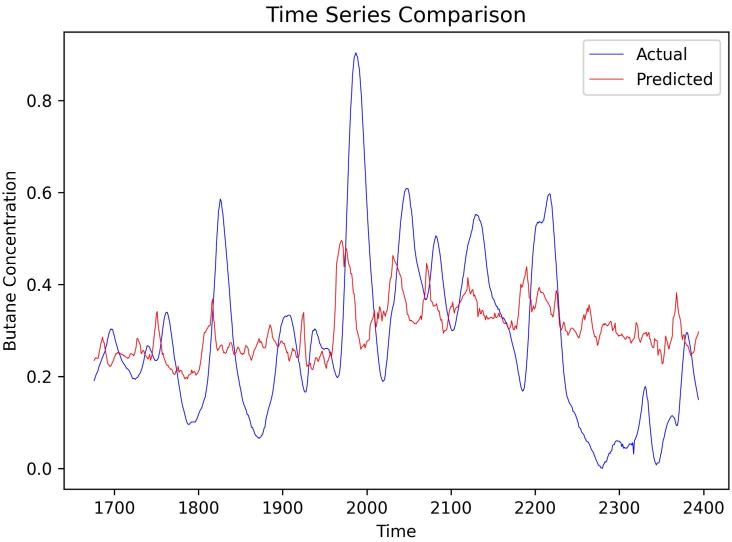
Actual vs Predicted - Linear & Ridge



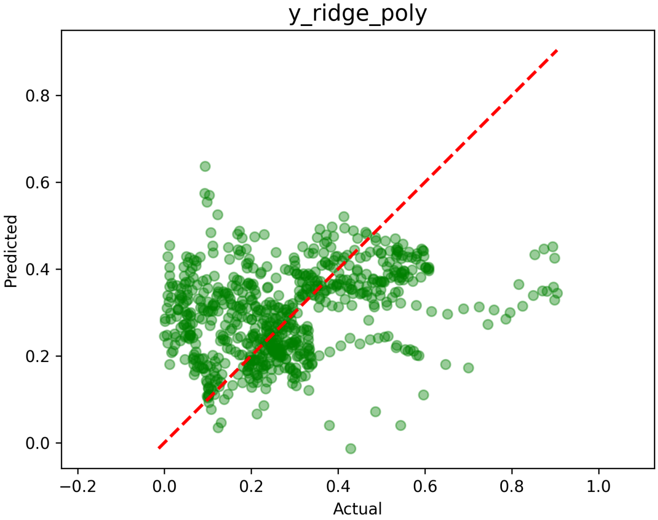
$R^2 = 0.1258$

RMSE $= 0.1678$

Time Series Comparision - Linear & Ridge



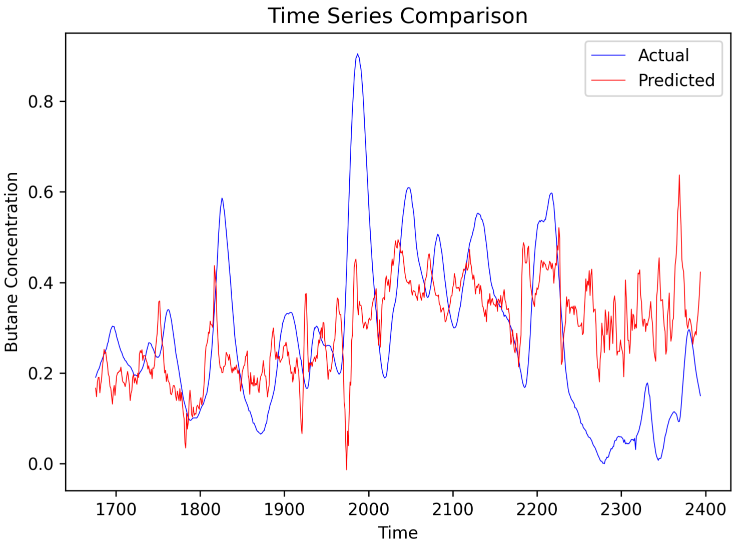
Actual vs Predicted - Polynomial(2) & (Ridge + Poly2)



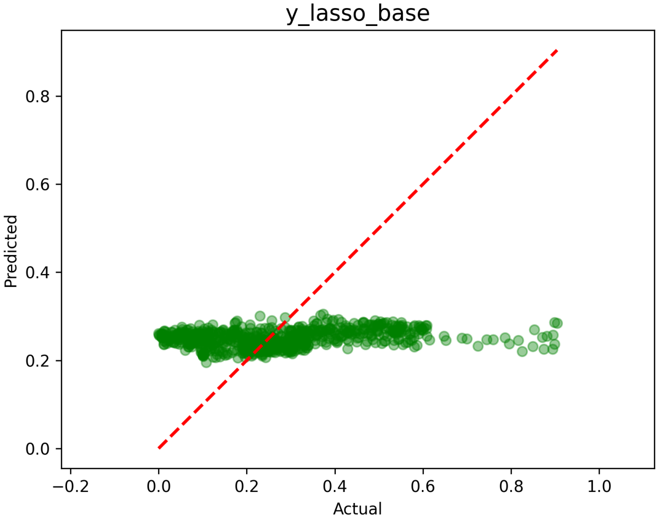
$R^2 = 0.0074$

$\text{RMSE} = 0.1788$

Time Series Comparision - Polynomial(2) & (Ridge + Poly2)



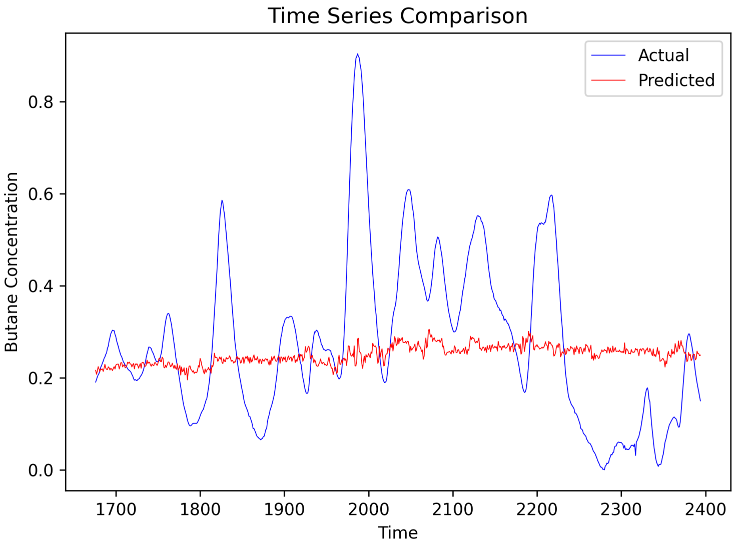
Actual vs Predicted - Lasso



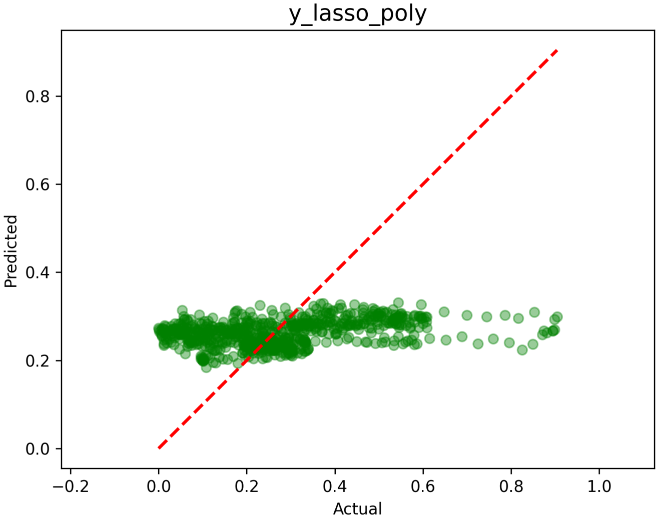
$R^2 = 0.0217$

$\mathrm{RMSE} = 0.1775$

Time Series Comparision - Lasso



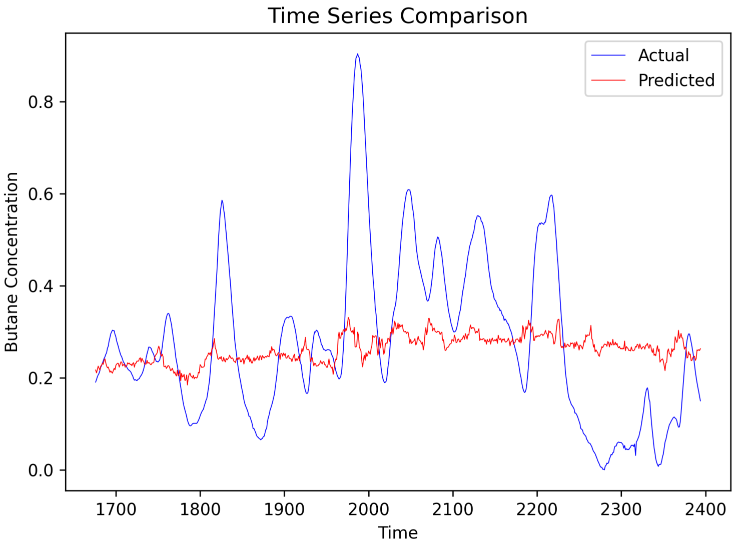
Actual vs Predicted - Lasso + Poly2



$R^2 = 0.078$

$\text{RMSE} = 0.1723$

Time Series Comparision - Lasso + Poly2



Best Performing Model: JITL + PLS

- Developed a **Just-In-Time (JIT) learning framework** for soft sensing
- Integrated **Partial Least Squares (PLS)** for handling multicollinearity
- Local models are built dynamically for each query point
- Improves prediction accuracy by focusing on **relevant neighborhood data**
- Suitable for **nonlinear and time-varying processes**

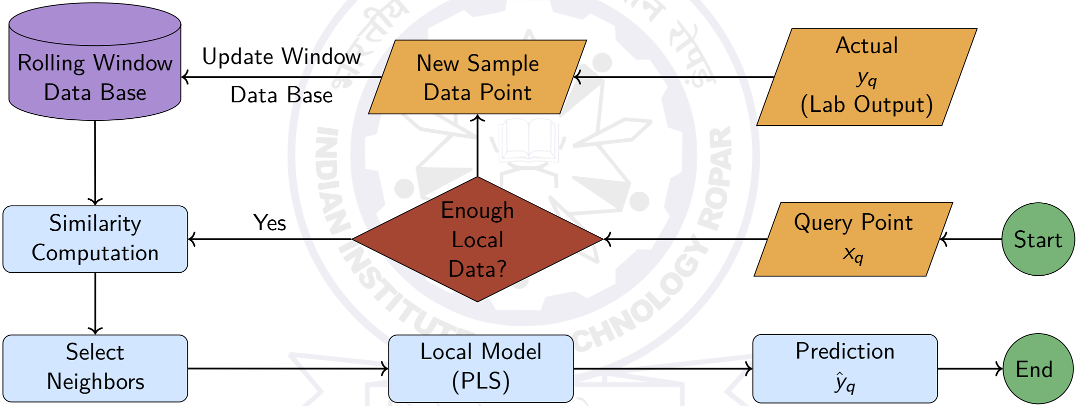
Enhancements: Polynomial & Rolling Window

- Introduced **Polynomial feature expansion**
  - Captures nonlinear relationships
  - Degree selected adaptively
- Implemented **Rolling Window strategy**
  - Uses most recent data for model building
  - Handles **concept drift** in dynamic systems
- Combined framework:

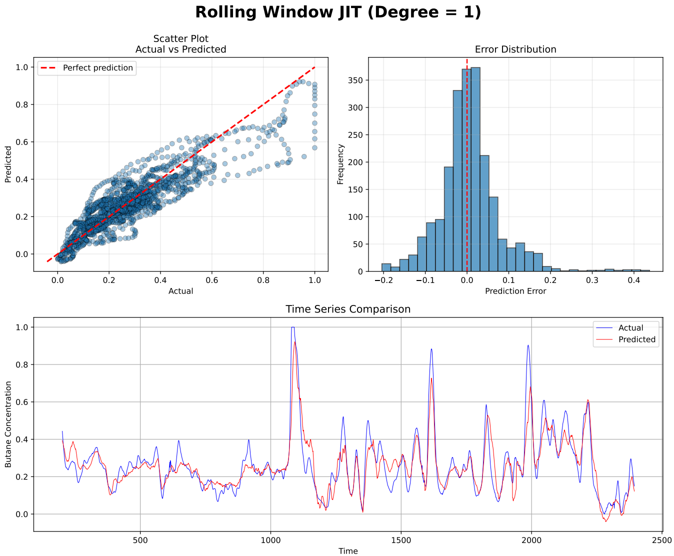
$$\text{JIT} + \text{PLS} + \text{Polynomial Features} + \text{Rolling Window}$$

- Results in improved **robustness and adaptability**

Rolling Window JITL Workflow



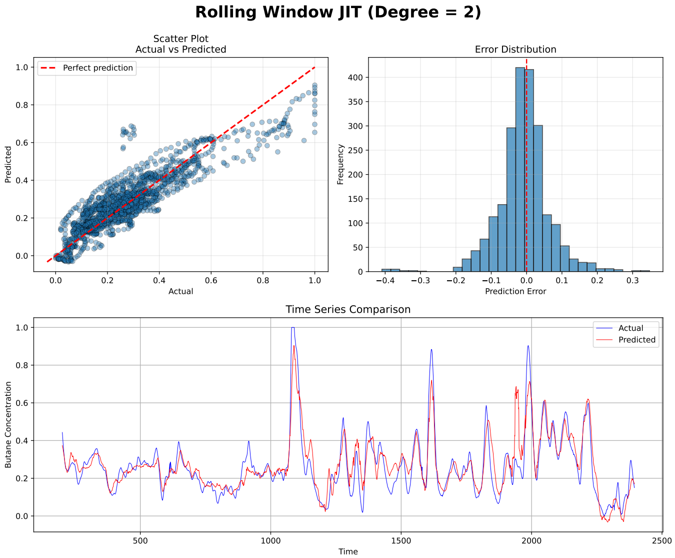
Best Performing Model: JITL + PLS



$R^2 = 0.7921$

$\text{RMSE} = 0.0737$

Best Performing Model: JITL + PLS



$R^2 = 0.7909$

$\text{RMSE} = 0.0739$

Model Performance Comparison

Model	R^2	RMSE
Linear & Ridge	0.1258	0.1678
Polynomial & (Ridge + Poly2)	0.0074	0.1788
Lasso	0.0217	0.1775
Lasso + Poly2	0.078	0.1723
JIT + PLS (Degree = 1)	0.7921	0.0737
JIT + PLS (Degree = 2)	0.7909	0.0739

Key Takeaways

- Polynomial & JITL handle nonlinearities effectively.
- Regularization (Lasso/Ridge) prevents overfitting.
- Median filtering improves robustness.
- Soft sensors reduce need for analyzers and enable real-time prediction.

Future Work

- Hyperparameter tuning for optimal performance.
- Cross-validation to avoid overfitting.
- Compare with advanced machine learning models.
- Explore real-time deployment feasibility.
- Test robustness against sensor noise.
- Adapt models to changing operating conditions.
- Evaluate long-term maintenance and retraining strategies.

References

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Thank You